SDG BLOCKCHAIN ACCELERATOR

Impact Measurement Framework

1. Project Information

- **Project Name:** KijaniMRV – Digital Measurement, Reporting & Verification Infrastructure for Climate Finance

- **Challenge & UNDP Office:** UNDP Ethiopia

- **Document Version:** v1.0

2. SDG Alignment

KijaniMRV directly supports transparent, verifiable climate action and sustainable development through digital MRV infrastructure.

SDG 13: Climate Action

- **Target:** Strengthen transparency and integrity of emissions reduction and removal reporting.
- **Indicator:** Tons of CO_2e verified, issued, and retired through the platform.

SDG 15: Life on Land

- **Target:** Promote sustainable land use, reforestation, and ecosystem restoration.
- **Indicator:** Hectares of land under verified reforestation or agroforestry monitoring.

SDG 12: Responsible Consumption and Production

- **Target:** Enable transparent and traceable carbon markets.
- **Indicator:** Number of verified carbon credit lots issued and publicly retired with provenance records.

3. Key Performance Indicators (KPIs)

KPI	Measurement Method	Target / Threshold	Notes
Verified emissions reductions/removals	Verified emissions reductions/removals	50,000 tCO_2e	Reported in tCO_2e
Carbon Credit Tokens Minted	On-chain ledger	50,000 credits	Each token represents 1 ton CO_2
Active users onboarded	Backend dashboard	10,000 users	Tracked via wallet addresses

4. Tracking Methods

On-chain Metrics

- Issuance and retirement transaction logs (Cardano testnet).
- Token state changes for carbon credit lots.
- Hash anchoring of MRV dossiers.
- DID-based authorization records for issuers and verifiers.

Off-chain Metrics

- MRV analytics outputs from satellite data and AI models.
- Mobile data capture logs (photos, plots, geotagged evidence).
- Verifier dashboard activity and decision logs.
- Buyer portal access and proof-of-retirement views.

5. Baseline vs Projected Outcomes

Metric	Baseline	Projected Outcome
Carbon credits issued	0	500,000/year
Carbon Credits Retired	0	≥80% of issued credits
Active users	0	10,000
Transaction success rate	N/A	≥95%
System Uptime	N/A	≥97%
End-to-end MRV cycle	N/A	≤14 days

6. Monitoring & Reporting Plan

- ***Frequency:***
  Monthly internal monitoring; quarterly external reporting.

- ***Responsible Parties:***
  Kitovu Product & Engineering Team; Verification Partners; PMU representatives.

- ***Format:***
  Dashboard analytics, on-chain logs, and structured PDF reports for stakeholders.

- ***Tools:***
  Cardano testnet logs, backend analytics dashboard, satellite analytics pipelines, and MRV reporting templates.

7. Risks & Mitigation

Risk	Mitigation
Delays in satellite data availability	Use of multiple satellite sources and historical baselines
Data inconsistencies between the field and the RS data	Verifier review workflows and exception flags
System performance issues during the pilot	Scoped pilot volumes and staged rollout
User friction with blockchain components	Wallet abstraction and backend-managed transactions
Verification disputes	Structured query/appeal workflows with audit trails